THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL UNIT TEST SET 1

Chapters 1–4: **Exploring Substances: Acidic, Basic and Neutral, Electricity: Circuits and their Components, The World of Metals and Non-metals, Changes Around Us: Physical and Chemical** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A student finds that a colourless liquid turns blue litmus red and does not affect red litmus. It also fizzes with baking soda. The liquid is: 1

(a) an acid **(b)** neutral **(c)** a salt **(d)** a base

2. A bulb rated 6 V is connected to a battery of three 1.5 V cells in series. The bulb will: 1

(a) glow dimly **(b)** fuse **(c)** glow normally **(d)** not glow at all

3. A shiny solid X conducts electricity, is hard, and its oxide turns red litmus blue. X is probably: 1

(a) phosphorus **(b)** sulphur **(c)** carbon **(d)** magnesium

4. A student heated a blue crystalline solid. It turned white and drops of liquid appeared on the cooler part of the tube. On adding water to the white powder, it turned blue and became warm. This indicates: 1

(a) a reversible chemical change **(b)** an irreversible change
(c) burning **(d)** a physical change

5. In an experiment, 20 mL of a base is neutralised by 20 mL of a certain acid of the same type. If 40 mL of the same base is to be neutralised, the acid needed is: 1

(a) 10 mL **(b)** 40 mL **(c)** 80 mL **(d)** 20 mL

6. A student claims: 'Pure water conducts electricity well'. How would you test and correct this? 1

(a) test with a tester; add salt to water to make it conduct
(b) heat it
(c) add sugar to make it conduct
(d) water is a good conductor

7. Which of these metals will give a vigorous reaction with dilute HCl? 1

(a) copper **(b)** gold **(c)** magnesium **(d)** silver

8. Which of these is the most reliable test to decide whether a change is chemical? 1

(a) change in shape **(b)** formation of a new substance
(c) change in position **(d)** change in size

9. A rainwater sample turns blue litmus red. The rain may be: 1

(a) acidic **(b)** basic **(c)** salty **(d)** neutral

10. A student sets up a tester with a cell, a bulb and two wires with free ends. The bulb glows when a steel spoon is placed across the ends. The spoon is: 1

(a) a fuse **(b)** a cell **(c)** an insulator **(d)** a conductor

11. Which metal is used to coat iron sheets to protect them from rusting? 1

(a) sulphur **(b)** wax **(c)** coal **(d)** zinc

12. A shop owner stores iron nails in a damp room. After some days, nails are covered with flaky reddish layer. The best advice to prevent this: 1

(a) store in water **(b)** store in vinegar

(c) store in dry place or oil them **(d)** store in salt water

13. A liquid turns turmeric red. When liquid Y is added, the colour changes back to yellow. Liquid Y is: 1

(a) an acid **(b)** water **(c)** sugar **(d)** a base

14. A student connects five 1.5 V cells in series to a 6 V bulb. The bulb will: 1

(a) glow very bright and may fuse **(b)** be dim

(c) not glow **(d)** glow normally

15. A yellow brittle solid Y burns with a blue flame and its oxide turns blue litmus red. Y is: 1

(a) sulphur **(b)** magnesium **(c)** iron **(d)** copper

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. When a few drops of lemon are added to a basic solution containing turmeric (red), the colour becomes yellow. Explain. 2

17. Explain why the filament of a bulb is made of a thin tungsten wire and not of copper. 2

18. Predict the products when zinc reacts with dilute sulphuric acid and write the equation. 2

19. A student weighs an iron nail before and after rusting and finds an increase in mass. Explain. 2

20. How can you distinguish between two colourless liquids, one acid and one base, with a single indicator? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Student X finds that 10 mL of acid is neutralised by 20 mL of base. How much acid neutralises 60 mL of base of the same strength? 3

22. A student connects 6 cells of 1.5 V each in series. Find the total voltage and say what happens to a 3 V bulb connected to it. 3

23. A student has an unknown sample. It is shiny, ductile, burns in air to a white powder that turns red litmus blue. Identify the class and give a probable example. 3

24. A student kept three iron nails: A in a closed jar with dry air and calcium chloride, B in boiled water with oil layer, C in a open beaker with tap water. Predict and justify results. 3

OR

Explain why we use different indicators for different solutions. 3

25. Why should heavy appliances be connected through thick wiring and not thin wiring? 3

OR

Explain with equations how metals and non-metals form acidic and basic solutions with water. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Design an experiment to find which of three antacid tablets neutralises the most acid. Describe variables and results. 5

OR

A string of ten small bulbs in series is connected to a battery and one bulb is fused. Explain what happens and how to find the faulty bulb. 5

27. You are given samples of sulphur, iron, carbon (graphite) and aluminium. Plan tests to identify each using properties and chemical reactions. 5

OR

Design an experiment to find out how three factors (air, water and salt) affect rusting, using six nails; describe the set-up, observations and conclusions. 5

END OF QUESTION PAPER Total: 50 Marks

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